SAVINGS GOAL PLANNING & FINANCIAL PROGRESS TRACKING SYSTEM

- A Java-based application to help users set savings goals, track income and expenses, and monitor financial progress automatically.

Java Project Report

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Abstract

- Many individuals struggle to save money due to lack of proper planning and monitoring. This system provides a structured, automated solution for goal setting and financial tracking.

- Users define savings goals (e.g., mobile purchase, education, travel), record income and expenses, and the system automatically calculates savings, remaining amount, and progress percentage.

GOAL SETTING

- Target amount + time period

AUTO CALCULATION

- Savings = Income - Expenses

PROGRESS DISPLAY

- Percentage toward goal

Objectives

SET FINANCIAL GOALS

- Define target amount and time period for each savings goal.

AUTOMATE CALCULATIONS

- Reduce manual errors by computing savings, remaining amount, and progress automatically.

TRACK PROGRESS

- Display savings progress as a percentage for clear financial visibility.

USER-FRIENDLY SYSTEM

- Menu-driven interface for easy financial management by any user.

Key Features

GOAL SETTING

- Set target amount and duration for any savings goal.

INCOME RECORDING

- Add salary, pocket money, or other earnings

EXPENSE TRACKING

- Log daily costs — food, travel, shopping, and more.

PROGRESS & SUMMARY

- View savings %, remaining amount, and full financial report.

System Modules

MENU DISPLAY

- Shows savings goals with target amounts and durations.

INCOME & EXPENSE ENTRY

- Accepts and stores user income and expense records.

SAVINGS CALCULATION

- Computes: Savings = Income − Expenses

REPORT GENERATION

- Displays goal, income, expenses, savings, remaining, and progress %.

EXIT

- Safely closes the application after all data is processed.

Methodology

PROBLEM ANALYSIS

- Identified gaps in manual financial tracking methods.

SYSTEM DESIGN

- Designed class structure, data flow, and user interface layout for optimal usability.

PROGRAM DEVELOPMENT

- Built using Java — classes, loops, conditionals, and OOP principles.

TESTING & VALIDATION

- Multiple test cases verified accurate savings and progress calculations.

SUCCESSFUL IMPLEMENTATION

- The Java application was successfully developed and deployed, meeting all initial requirements.

Algorithm

CORE FORMULA

Saving= Income - Expenses

Remaining = goal- Saving

$$\text{Process}(\%) = \frac{\text{Saving}}{\text{Goal}} \times 100$$

01

DISPLAY MENU

- Set Goal / Add Income / Add Expense / View Summary / Exit

02

ENTER DATA

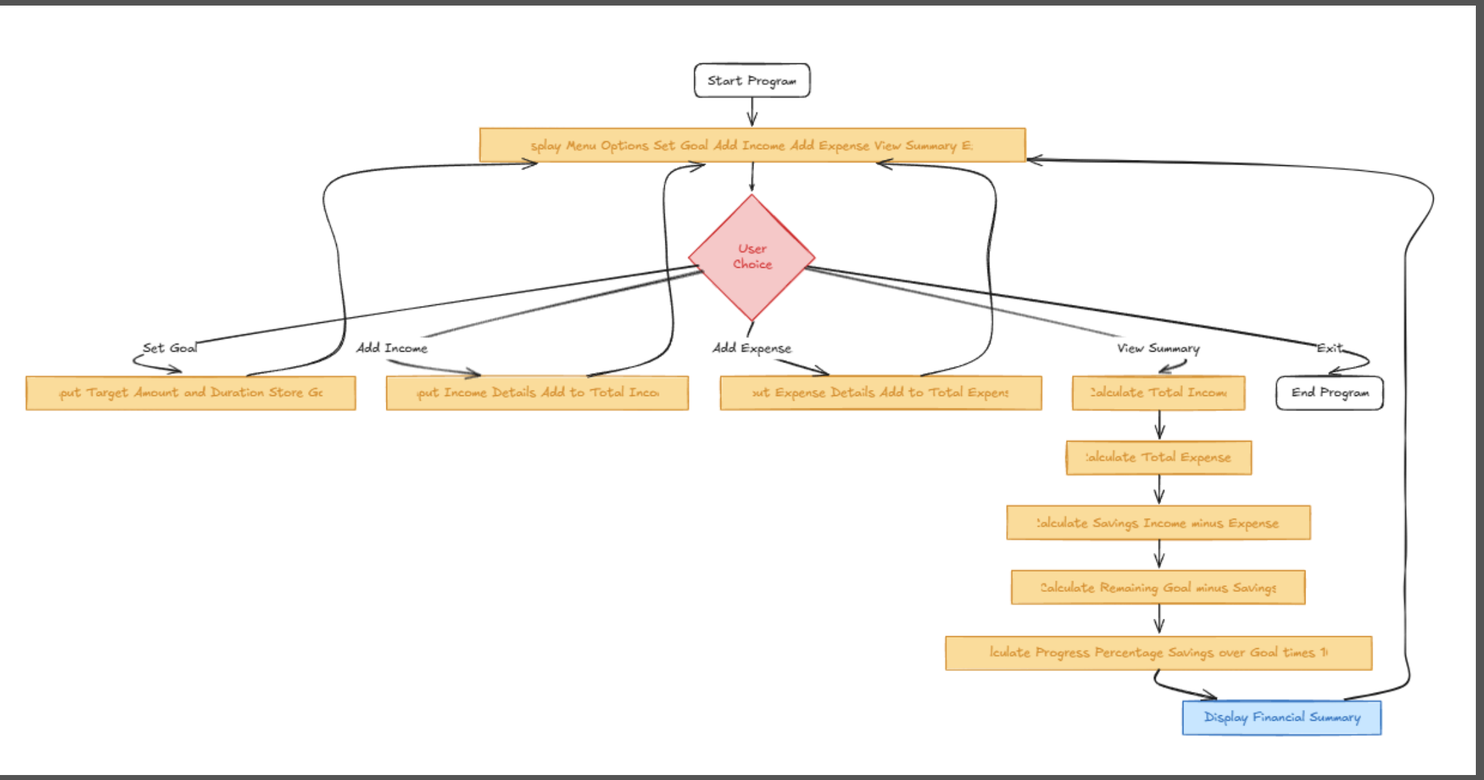
- User inputs goal amount, income, and expense values.

03

CALCULATE & DISPLAY

- System computes savings, remaining amount, and progress %.

System Flowchart



- The flowchart illustrates the complete program flow — from displaying the menu to processing each user choice (Set Goal, Add Income, Add Expense, View Summary, Exit) and looping back until the user exits

Source Code & Sample Output

```
import java.util.Scanner;
class SavingsSystem {
 public static void main(String[] args) {
  Scanner sc = new Scanner(System.in);
  double goalAmount = 0, totalIncome = 0,
      totalExpense = 0;
  int choice;
  do {
   System.out.println("1. Set Savings Goal");
   System.out.println("2. Add Income");
   System.out.println("3. Add Expense");
   System.out.println("4. View Summary");
   System.out.println("5. Exit");
   choice = sc.nextInt();
   switch (choice) {
    case 1: goalAmount = sc.nextDouble(); break;
    case 2: totalIncome += sc.nextDouble(); break;
    case 3: totalExpense += sc.nextDouble(); break;
    case 4:
     double savings = totalIncome - totalExpense;
     double remaining = goalAmount - savings;
     double progress = (savings/goalAmount)*100;
     System.out.println("Savings: ₹" + savings);
     System.out.println("Progress: " + progress +
"%");
    }
  } while (choice != 5);
 }
}
```

SAMPLE OUTPUT:

```
----- SAVINGS REPORT -----
Goal Amount:   ₹500.0
Total Income:  ₹200.0
Total Expense: ₹50.0
Savings:       ₹150.0
Remaining:     ₹350.0
Progress:      30.0%
```

TESTED CASES :

goal	income	expenses	process
RS: 10,000	RS: 5,000	RS: 2,000	30%
RS: 20,000	RS: 15,000	RS: 5,000	50%
RS.5,000	RS.5,000	0	100%

Conclusion & References

CONCLUSION

- The system successfully automates income, expense, and savings tracking — reducing errors and improving financial awareness. It can be further enhanced with database integration, GUI, mobile support, and automated notifications.

REFERENCES

- Herbert Schildt – Java: The Complete Reference
- oracle.com/java
- geeksforgeeks.org
- w3schools.com ·